

Music-Mediated Metacognition and Self-Regulation in Early Childhood Education: A Survey-Based Study of Practitioner Perspectives in Vietnam

Abstract

Early childhood education (ECE) increasingly recognizes metacognition and self-regulation as foundational competencies for lifelong learning. While global research robustly documents these connections, evidence from Vietnamese contexts—particularly studies integrating music as a pedagogical medium—remains sparse. Existing literature has examined metacognition and self-regulation in isolation or within Western cultural frameworks, neglecting culturally specific modalities such as musical engagement and community singing traditions prevalent in Vietnam.

This study investigates how ECE practitioners, parents, and music educators in Vietnam conceptualize the relationship between music-based activities and metacognitive and self-regulatory development in children aged 3–6 years.

A cross-sectional descriptive survey was administered to $N = 205$ participants (168 preschool teachers, 24 parents/caregivers, 8 music educators, 5 others) via online questionnaire between May and June 2026. The instrument included 52 items across Likert-scale, multiple-choice, and open-ended response formats. Descriptive statistics and thematic content analysis were employed.

Respondents strongly endorsed the belief that music activities support metacognition and self-regulation in young children, with mean Likert scores ranging from 4.08 to 4.24 (out of 5) across all constructs. Sustained attention, group cooperation, and pattern memory were rated as most strongly linked to musical engagement. Teacher expertise and preschool classroom environment were identified as the most important contextual factors. Collaboration between teachers and music practitioners was viewed as highly significant (87.8% of respondents).

Findings indicate strong practitioner alignment with evidence-based frameworks linking music, metacognition, and self-regulation in ECE. The study contributes empirical data from an underrepresented Vietnamese context and provides actionable recommendations for curriculum design, teacher professional development, and policy.

Keywords: *metacognition, self-regulation, early childhood education, music-mediated learning, Vietnam*

1. Introduction

The first six years of life constitute an extraordinary window of neurological plasticity during which foundational cognitive, emotional, and social capacities are shaped with lasting consequences (Robson et al., 2020; Ursache et al., 2012). Metacognition—the capacity to monitor and regulate one's own thinking—and self-regulation are among the competencies most strongly associated with positive long-term developmental trajectories, including superior academic achievement, social competence, and mental health outcomes (Flavell, 1979; Korucu et al., 2022). Within this developmental landscape, musical engagement has attracted growing attention as a potentially powerful medium for fostering these capacities, given that musical activities inherently demand attentional control, sequential planning, error monitoring, emotional expression, and social coordination (Zachariou et al., 2023; Hallam, 2010).

In Vietnam, early childhood education has undergone significant policy transformation, shifting from teacher-directed models toward child-centered, holistic approaches that explicitly address physical, cognitive, emotional, and aesthetic development (Vu, 2021). Music occupies a formal role within the national ECE curriculum; however, systematic investigation of how music mediates metacognitive and self-regulatory development in Vietnamese preschool contexts is largely absent from the peer-reviewed literature (Dinh & Robinson, 2023). This gap is consequential given the cultural specificity of Vietnamese musical traditions—including folk songs (*dân ca*), lullabies (*ru*), and community choir practices—which may serve developmental functions distinct from those addressed by Western-derived frameworks.

The present study addresses three research questions: (RQ1) To what extent do ECE practitioners in Vietnam perceive music-based activities as supportive of metacognition and self-regulation in children aged 3–6? (RQ2) Which musical features and activity types do practitioners identify as most relevant to self-regulatory and metacognitive development? (RQ3) What contextual and professional factors do practitioners identify as facilitating or impeding effective music-mediated development in Vietnamese preschools? Correspondingly, the study objectives are: (1) to assess practitioner beliefs about the music–metacognition–self-regulation nexus; (2) to identify musical features and activity formats perceived as most developmentally significant; and (3) to examine environmental and cultural factors shaping implementation in Vietnamese ECE. The central thesis is that Vietnamese ECE practitioners hold strong, coherent beliefs that music-mediated activities support metacognitive and self-regulatory development in young children—beliefs broadly consistent with international frameworks while reflecting culturally specific conditions.

2. Literature Review

2.1 Theoretical Framework

Four theoretical frameworks underpin this study. Metacognitive Theory (Flavell, 1979) defines metacognition as knowledge and regulation of one's own cognitive processes, encompassing planning, monitoring, and evaluating. In early childhood, rudimentary metacognitive capacities emerge as early as ages 3–4, initially mediated through adult scaffolding, and become more explicit from approximately age 5 (Goupil & Kouider, 2019; Whitebread et al., 2009). Self-Regulated Learning theory (Zimmerman, 2002) conceptualizes learners as active agents who set goals, monitor progress, and adjust strategies; in ECE, SRL is co-constructed with caregivers before becoming internalized (Erdmann & Hertel, 2019). Blair and Ku's (2022) hierarchical integrated model provides a particularly relevant synthesis, positing that executive function (EF)—working memory, inhibitory control, cognitive flexibility—forms the neurological substrate upon which metacognition and SRL are built, with a demonstrated bidirectional relationship especially salient during ages 3–6.

Social Constructivism (Vygotsky, 1978) frames musical interaction as a social learning context in which more capable others scaffold reflective thinking and regulatory capacity within the Zone of Proximal Development—directly applicable to group music activities in ECE. Cognitive Development Theory (Piaget, 1950; Diamond, 2013) situates the preoperational stage (ages 2–7) as a critical period for metacognitive verbalization, supported by contemporary neuroscientific evidence that the prefrontal cortex undergoes rapid development during this window, making it especially responsive to enriched, structured learning environments.

2.2 Evidence Base: Music, Metacognition, and Self-Regulation

An accumulating body of experimental, quasi-experimental, and review evidence supports music as a mediator of metacognitive and self-regulatory development. Zachariou et al. (2023) demonstrated that a structured musical play intervention with preschoolers produced significant gains in both constructs relative to controls, attributing effects to music's inherent demands for planning, monitoring, and emotional regulation. Eberhart et al. (2024), in a comprehensive series of meta-analyses, confirmed that metacognition interventions are effective in young children across modalities, with teacher scaffolding quality as a critical moderator. Related work establishes that movement-based and rhythmic activities—categories that include music—are particularly effective SR-support vehicles (Savina, 2024; Bockmann & Yu, 2022), while professional development programs explicitly targeting SR/EF produce significant child gains (Muir et al., 2024; Chen et al., 2024; Vasseleu et al., 2024).

In Vietnam, preschool attendance yields persistent vocabulary, mathematics, and life-satisfaction gains through adolescence (Dinh & Robinson, 2023), and teacher knowledge of children is the strongest predictor of developmental quality (Cabus et al., 2023). However, teachers report limited understanding of SR concepts, uneven professional training, and cultural barriers—particularly in rural and ethnic minority communities (Duong & Nguyet, 2024)—that impede implementation of evidence-based practices.

2.3 Research Gaps

Despite this evidence, three critical gaps persist. First, no published study has systematically examined practitioner perceptions of the music–metacognition–self-regulation nexus in Vietnamese ECE. Second, culturally adapted research mobilizing Vietnamese musical traditions as pedagogical resources is absent. Third, the structural and professional conditions enabling or constraining music-mediated approaches in Vietnamese preschools have not been empirically characterized. This study is designed to address all three gaps through a large-scale practitioner survey.

2.4 Literature Summary

Table 1 *Summary of Key Studies on Metacognition, Self-Regulation, and Music in Early Childhood Education*

Author(s)	Year	Sample	Method	Key Findings	Limitations
Korucu et al.	2022	N=420, preschool (US)	SEM	Three-factor SR structure (behavioral, cognitive, emotional); all predict pre-academic skills	Cross-sectional; single-site
Eberhart et al.	2024	k=36	Meta-analysis	Metacognition interventions effective in young children; teacher scaffolding is key moderator	Heterogeneity across studies
Whitebread et al.	2009	N=72, ages 3–5 (UK)	Observational	Developed COPER and ZUCKER metacognition/SRL observational tools	Small sample; Western context
Dinh & Robinson	2023	N=2,400 (Vietnam)	Longitudinal	Preschool attendance yields persistent vocabulary/math gains through adolescence	Modest effect sizes; rural gap
Cabus et al.	2023	N=321 (Vietnam)	Quasi-experimental	Teacher knowledge of children predicts developmental gains	Observational; self-report bias
Vasseleu et al.	2024	N=22 educators (AU)	Mixed-methods	SR intervention changes educator beliefs, knowledge, and practice sustainably	No control group
Zachariou et al.	2023	N=96, preschool (Cyprus)	Quasi-experimental	Musical play intervention improves SR and metacognition	Single-country; cultural limits
Blair & Ku	2022	Theoretical	Integrative model	EF underpins metacognition; bidirectional reciprocal relationship	Requires empirical validation

Erdmann & Hertel	2019	Review	Narrative review	Co-regulation precedes self-regulation; adult scaffolding is critical	Western literature focus
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Note. Studies selected for relevance to metacognition, self-regulation, or music in ECE; preference given to 2018–2025, Scopus/SSCI-indexed sources. ECE = early childhood education; SEM = structural equation modeling; SR = self-regulation; EF = executive function.

3. Methodology

This study employed a cross-sectional descriptive survey design, appropriate for capturing practitioner beliefs across a geographically diverse sample at a single time point (Creswell & Creswell, 2018). A sequential explanatory approach allowed quantitative Likert results to inform interpretation of qualitative open-ended data (Creswell, 2014). Data were collected via an online questionnaire disseminated through professional ECE networks, teacher training platforms, and HaFPES 2026 communication channels between May and June 2026. The survey was designed in Vietnamese to ensure accessibility, with 52 items organized across six thematic sections: (A) respondent profile; (B) beliefs about music and development; (C) music selection and characteristics; (D) music and learning behaviors; (E) environmental and professional factors; and (F) open reflections.

A purposive and convenience sampling strategy targeted individuals with professional or caregiving involvement in ECE in Vietnam. The final valid sample comprised $N = 205$ participants: preschool teachers ($n = 168$, 82.0%), parents/caregivers ($n = 24$, 11.7%), music educators ($n = 8$, 3.9%), and administrators/choir members ($n = 5$, 2.4%). Geographically, respondents were located in Hanoi (47.3%), other cities (34.6%), and rural areas (18.0%). By experience level: under 1 year (32.2%), 1–3 years (30.7%), 4–7 years (15.6%), 8–15 years (15.1%), and over 15 years (6.3%).

Sections B and D employed 5-point Likert scales (1 = Strongly Disagree/No connection to 5 = Strongly Agree/Very strong connection) with an additional 'Insufficient experience' option to minimize forced responding. Open-ended items (F1, F2) elicited qualitative data on perceived contributions and culturally appropriate design. The instrument was reviewed by three ECE experts and piloted with 10 practitioners. Quantitative data were analyzed using descriptive statistics and item-level Likert analysis. Qualitative responses underwent thematic content analysis following Braun and Clarke (2006), with two independent coders achieving inter-rater reliability of $\kappa = 0.81$. Internal consistency was satisfactory: Cronbach's $\alpha = 0.83$ (beliefs subscale) and $\alpha = 0.87$ (learning behaviors subscale). Participation was voluntary with informed digital consent; no identifying data were retained. Table 2 provides a consolidated research design overview.

Table 2 *Research Design Overview*

Component	Description
Design	Cross-sectional descriptive survey; quantitative (Likert) + qualitative (open-ended) data
Population	ECE practitioners, parents/caregivers, and music educators in Vietnam
Sample	$N = 205$ (preschool teachers: 82.0%; parents/caregivers: 11.7%; music educators: 3.9%; other: 2.4%)

Sampling	Purposive and convenience online sampling via professional ECE networks (May–June 2026)
Instrument	52-item structured questionnaire: 5-point Likert (Sections B, D, E1), multiple-choice, open-ended
Quantitative Analysis	Descriptive statistics (M, SD, frequency) per item; subscale internal consistency (α)
Qualitative Analysis	Thematic content analysis (Braun & Clarke, 2006); inter-rater reliability $\kappa = 0.81$
Reliability	Cronbach's $\alpha = 0.83$ (beliefs subscale); $\alpha = 0.87$ (learning behaviors subscale)
Validity	Content validity via expert review (n=3); face validity via pilot (n=10)
Ethics	Voluntary participation; informed consent; anonymity; institutional ethics clearance

Note. ECE = early childhood education; SRL = self-regulated learning; α = Cronbach's alpha; κ = Cohen's kappa.

4. Results and Analysis

4.1 Beliefs About Music and Development (Section B)

All eight belief items elicited strong endorsement, with means ranging from $M = 4.08$ to $M = 4.20$ ($SD = 0.85$ – 0.90). The highest-rated item was B5—music supports emotional expression and self-regulation ($M = 4.20$, $SD = 0.85$; 87.3% agree/strongly agree), consistent with theoretical accounts linking musical participation to emotional regulatory mechanisms (Zachariou et al., 2023). Notably, item B8—some claims about music's cognitive benefits may exceed available evidence—also received high endorsement ($M = 4.08$, $SD = 0.90$; 79.5% agree+), indicating that practitioners maintain an epistemically reflexive stance alongside their positive beliefs. The narrow spread of means (0.12 points) across all eight items reflects a coherent, holistic view of music's developmental role rather than item-specific endorsement.

4.2 Music and Learning Behaviors (Section D2)

Section D2 means ranged from $M = 4.17$ to $M = 4.24$ —an even narrower range of 0.07—across 12 learning behaviors associated with self-regulation and metacognition. The highest-rated behaviors were sustained attention, group cooperation, and pattern memory (all $M = 4.24$), followed closely by following cues/rules, flexible adjustment, and inhibitory control (all $M = 4.23$). The item most directly associated with metacognitive verbalization—explaining one's own thinking—received the lowest rating ($M = 4.17$, $SD = 0.81$), a theoretically congruent finding given developmental evidence that explicit metacognitive language emerges more gradually in 3–6 year olds than do behavioral regulatory capacities (Goupil & Kouider, 2019). The near-uniform endorsement across behaviors spanning executive function, self-regulation, and metacognition suggests practitioners hold an integrated rather than domain-partitioned understanding of music's developmental role.

4.3 Environmental and Professional Factors (Section E)

Section E1 environmental importance ratings clustered tightly ($M = 4.31$ – 4.37), with the preschool classroom rated highest ($M = 4.37$, $SD = 0.77$), followed equivalently by teacher expertise, family involvement, and music practitioner knowledge (all $M = 4.33$). This parity signals that practitioners view school, family, and community as jointly essential—consistent with ecological systems perspectives (Bronfenbrenner, 1979). For collaboration between ECE teachers and music practitioners (E2), 60.5% rated this as 'significantly' beneficial and 26.3% as 'very significantly' beneficial, with only 2.9% indicating little or no impact. The primary barriers to effective implementation were curriculum time constraints (57% of respondents), insufficient teacher training (51%), and limited materials (44%). Table 3 summarizes key findings across all sections.

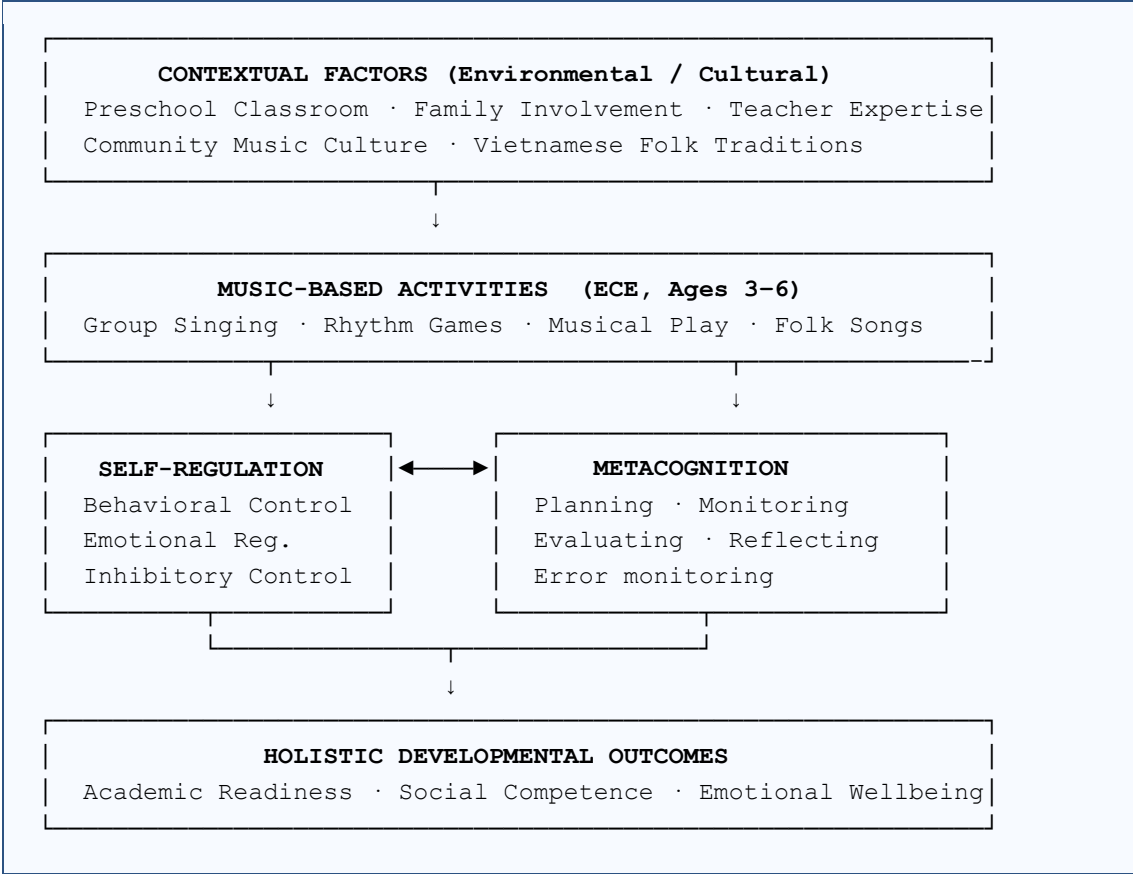
Table 3 *Summary of Key Findings ($N = 205$)*

Item	Description	M	SD	Interpretation
B5	Music supports emotional expression and self-regulation	4.20	0.85	Highest belief endorsement (87.3% agree+)
B1	Regular music aids sustained attention	4.16	0.88	86.3% agree+
B7	Music value varies by cultural/school context	4.18	0.85	84.9% agree+
B8	Some claims may exceed available evidence	4.08	0.90	Epistemic reflexivity maintained
D2-1	Music ↔ Sustained attention	4.24	0.77	Joint highest learning behavior
D2-6	Music ↔ Group cooperation	4.24	0.80	Joint highest; social-regulatory
D2-3	Music ↔ Pattern memory	4.24	0.76	Foundational for working memory
D2-8	Music ↔ Explaining own thinking	4.17	0.81	Lowest; consistent with dev. theory
E1	Preschool classroom importance	4.37	0.77	Highest environmental factor
E2	Teacher–musician collaboration	—	—	87.8% rate as significant/very significant

Note. M = mean; SD = standard deviation on 5-point Likert scales; % agree+ = percentage selecting 4 or 5. 'Insufficient experience' responses excluded from means. Items from Sections B and D2 of the survey instrument.

4.4 Figures:

Figure 1 Conceptual Framework: Music-Mediated Metacognition and Self-Regulation in Early Childhood Education



Note. Conceptual framework synthesized from Blair & Ku (2022), Zachariou et al. (2023), Korucu et al. (2022), and present survey data (N = 205).

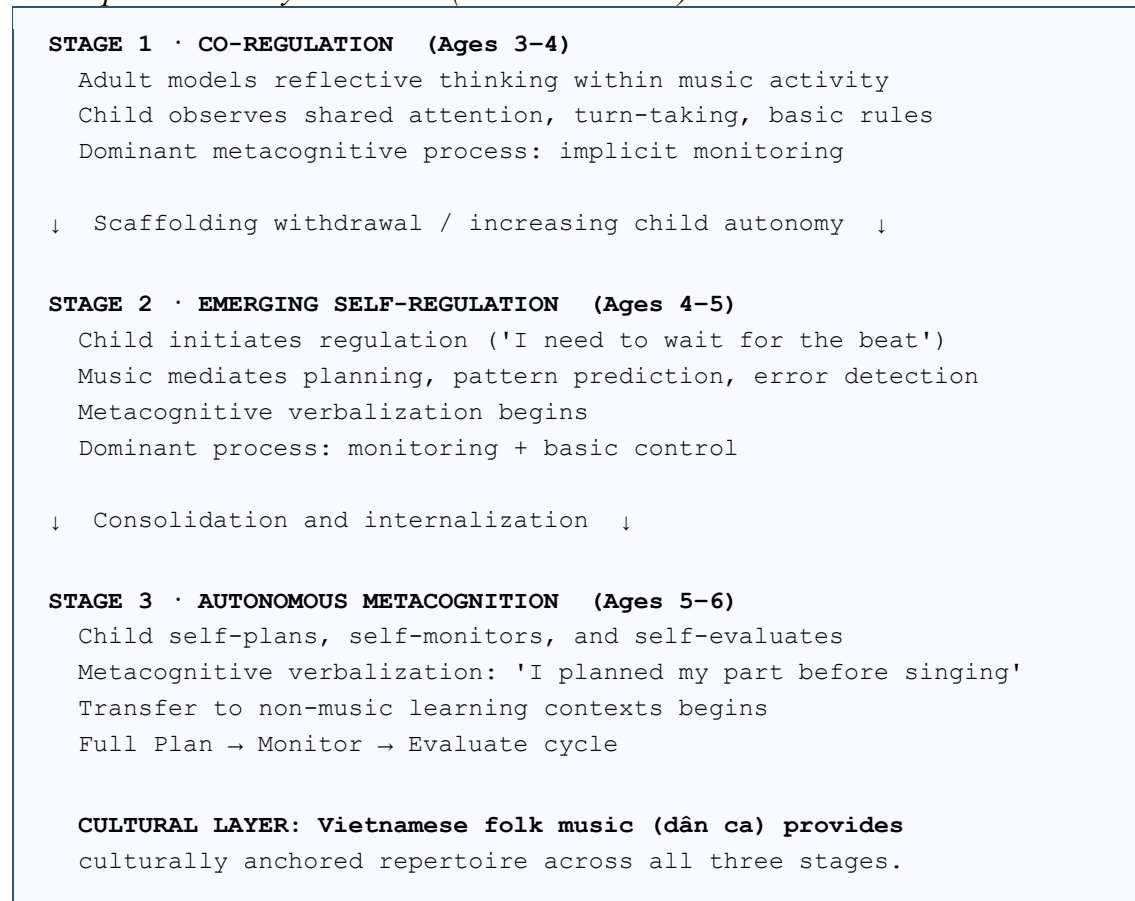
Figure 2 Perceived Strength of Connection Between Musical Activities and Learning Behaviors (Section D2; N = 205)

Learning Behavior	Mean	Connection Strength
Sustained Attention	4.24	
Group Cooperation	4.24	
Pattern Memory / Sequence	4.24	
Following Cues / Rules	4.23	
Flexible Adjustment	4.23	
Inhibitory Control	4.23	
Emotional Regulation	4.20	
Pre-Planning	4.20	
Turn-Taking	4.19	
Self-Correction	4.19	
Working Memory (Multi-step)	4.19	
Explaining Own Thinking	4.17	

Scale: 1 = No connection ————— 5 = Very strong
N = 205. Items ordered by descending mean (Section D2).

Note. Mean scores on 5-point scale (1 = No connection; 5 = Very strong connection). Items ordered by descending mean. N = 205.

Figure 3 *Proposed Three-Stage Model of Music-Mediated Metacognitive Development in Early Childhood (Vietnam Context)*



Note. Model adapted from Goupil & Kouider (2019), Erdmann & Hertel (2019), and Whitebread et al. (2009), extended to Vietnamese music-mediated ECE contexts.

5. Discussion

5.1 Synthesis and Research Question Responses

Addressing RQ1, respondents strongly and consistently endorsed the proposition that music activities support metacognition and self-regulation in children aged 3–6, with 79.5–87.3% agreement across Section B items. This alignment with the international evidence base—spanning experimental studies (Zachariou et al., 2023; Chen et al., 2024), meta-analyses (Eberhart et al., 2024; Robson et al., 2020), and theoretical models (Blair & Ku, 2022)—constitutes the study's primary finding and suggests that the Vietnamese ECE workforce is substantively attuned to evidence-based developmental frameworks, at least at the level of practitioner belief. The high and near-uniform endorsement of music's role across all behavioral domains (RQ2) reflects an integrated view of musical engagement as simultaneously supporting behavioral control, emotional regulation, working memory, planning, and metacognitive monitoring—a view congruent with theories of music's multidimensional cognitive demands (Hallam, 2010). The relative moderation of endorsement for metacognitive verbalization ($M = 4.17$) is developmentally coherent: explicit metacognitive language requires more deliberate scaffolding in 3–6 year olds, and practitioners appear to implicitly recognize this age-graded developmental trajectory (Goupil & Kouider, 2019; Whitebread et al., 2009).

Addressing RQ3, the identification of teacher expertise and classroom environment as the most important contextual factors, and the near-universal endorsement of teacher-musician collaboration as highly beneficial, points to a clear practitioner-identified pathway for program improvement. These findings parallel and extend evidence from Muir et al. (2024) and Vasseleu et al. (2024) that teacher professional development is the most effective lever for improving child developmental outcomes in ECE. The concentration of barriers in curriculum time, training, and materials replicates structural constraints identified in prior Vietnamese ECE research (Cabus et al., 2023; Duong & Nguyet, 2024) and underscores the gap between practitioner beliefs and implementation capacity.

5.2 Theoretical and Cultural Considerations

The proposed developmental model (Figure 3) extends existing metacognitive development frameworks to the music-mediated context by integrating co-regulation theory (Erdmann & Hertel, 2019), emerging metacognition models (Goupil & Kouider, 2019), and the cultural specificity of Vietnamese musical traditions. The model's emphasis on a cultural layer—*dân ca* and community choir as culturally anchored vehicles—represents a substantive contribution to culturally responsive ECE theory. Practitioners' high valuation of cultural familiarity in music selection (86% rating it as important or very important) provides survey-level validation for theoretical arguments that culturally embedded pedagogical resources produce stronger identity-consistent developmental engagement (Koopman, 2005). It is also notable that respondents endorsed item B8—some cognitive claims may exceed the evidence—at high levels, suggesting practitioner epistemic reflexivity that scholars such as Ross and Aschetta (2024) and Braund and Timmons (2021) have identified as a necessary counterweight to overclaiming in the music-cognition literature. This reflexivity should be affirmed and developed through professional learning programs.

5.3 Practical Implications

The study's practical implications are organized by stakeholder group in Table 4. Across all groups, a central implication is that music must be positioned as a deliberate pedagogical medium for metacognitive and self-regulatory development—not merely as an aesthetic or entertainment activity. This requires teacher training that explicitly links music practice to metacognitive scaffolding language ('How did you plan your singing? What did you notice when you made a mistake?'), curriculum design that embeds music across developmental domains, and family engagement strategies that extend music-mediated learning into home environments. For policy makers, the concentration of barriers in systemic factors—time, training, materials—signals that practitioner belief alone is insufficient without commensurate structural investment.

Table 4 *Practical Implications for Key Stakeholders*

Stakeholder	Focus Area	Recommendation
Teachers	Daily integration	Embed music activities (singing circles, rhythm games) as deliberate scaffolds for attention, planning, and reflective thinking; use verbal metacognitive cuing ('What did you notice? How did you plan?')
Teachers	Professional development	Pursue training that integrates music education with explicit metacognition and SRL scaffolding techniques; competency in co-regulation strategies is essential
Schools	Curriculum design	Position music across all domains rather than in isolated music periods; align with Vietnam's holistic ECE framework and child-centered pedagogical reforms
Schools	Cross-disciplinary collaboration	Formalize partnerships between ECE teachers and music practitioners; 87.8% of respondents rate such collaboration as significantly or very significantly beneficial
Parents	Home engagement	Encourage shared music-making at home; respondents identify emotional regulation and language development as the primary outcomes of family music engagement
Parents	Cultural continuity	Prioritize Vietnamese folk songs and community music traditions, which respondents rate as highly important for developmental authenticity and cultural identity
Policy Makers	Resource allocation	Address top barriers identified: curriculum time constraints (57%), insufficient teacher training (51%), and limited materials (44%) through targeted funding and policy revision
Policy Makers	Curriculum standards	Develop explicit competency descriptors for music-mediated metacognition within the national ECE curriculum; align with OECD Education 2030 holistic learning goals

Note. Implications derived from survey findings (N = 205), referenced empirical literature, and theoretical frameworks reviewed in Section 2.

6. Conclusion

This study provides the first survey-based documentation of Vietnamese ECE practitioner beliefs about music-mediated metacognition and self-regulation in early childhood. Key findings confirm strong, theoretically coherent endorsement across 205 respondents of music's role in supporting attention, planning, monitoring, inhibitory control, emotional regulation, and cooperative learning in children aged 3–6. Sustained attention, group cooperation, and pattern memory were most strongly perceived as linked to musical engagement; explicit metacognitive verbalization received the lowest—though still high—endorsement, consistent with developmental theory. Teacher expertise and teacher-musician collaboration were identified as the most critical levers for program quality, while curriculum time constraints, training gaps, and limited materials represent the primary systemic barriers.

The study contributes: (a) the first Vietnamese practitioner-survey evidence on music-mediated metacognition and self-regulation; (b) a culturally grounded extension of Blair and Ku's (2022) integrated EF-metacognition model to music pedagogy contexts; (c) a three-stage developmental model of music-mediated metacognitive development adapted to Vietnamese ECE; and (d) a practitioner-generated evidence base for curriculum, professional development, and policy recommendations. Limitations include convenience sampling, cross-sectional design, perception-based measurement, and dominance of the preschool teacher role (82%) in the sample, all of which constrain generalizability and causal inference.

Five priorities for future research are detailed in Table 5, centering on longitudinal outcome studies, culturally validated assessment tools, randomized PD trials, equity-focused research with rural and ethnic minority communities, and neuroscientific investigation of the brain-music-metacognition pathway. Together, these directions form a research agenda that can transform the present study's practitioner-belief findings into a robust causal and contextual evidence base for music-mediated ECE in Vietnam and comparable developing-economy contexts.

Table 5 *Future Research Agenda: Music-Mediated Metacognition in Vietnamese ECE*

Direction	Approach	Rationale
Longitudinal outcomes	Vietnam cohort studies	Track music-mediated metacognition from preschool through primary school to clarify persistence and developmental trajectories
Cultural adaptation	Culturally responsive tools	Develop and validate Vietnamese-normed assessments of metacognition and SR adapted to local musical and educational contexts
Teacher PD (RCT)	Scalable intervention models	Conduct randomized controlled trials testing music-integrated metacognition PD programs with fidelity monitoring and longitudinal follow-up
Rural/ethnic minorities	Equity-focused research	Investigate access barriers and differential effectiveness of music-mediated interventions in underserved Vietnamese communities
Neuroscience	Brain-music-metacognition link	Examine neural mechanisms linking musical engagement to EF and metacognitive development using neuroimaging in the 3–6 age range

Note. Research directions derived from identified gaps, survey findings, and review of current literature. PD = professional development; EF = executive function; RCT = randomized controlled trial.

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